

AP Statistics Class 11: Random Variables & Expected Value

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This class pack covers the entire AP Statistics random-variables unit: 12 concept posters with full theory notes and 5 practice questions each (60 total). Use it as your reference and worksheet — solutions are in the separate teacher answer key.

1. What Is a Random Variable?

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What Is a Random Variable?

A **random variable** is a numerical value determined by the outcome of a random process.

**Example:**
Roll a die and let X = the number rolled.
Possible values:
1, 2, 3, 4, 5, 6

**Example:**
Flip 3 coins and let X = the number of heads.
Possible values:
0, 1, 2, 3

It is called **random** because the value depends on chance.

CLASS 11 • AP Statistics

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Definition

A **random variable** is a numerical value determined by the outcome of a random process. We usually write it with a capital letter — X , Y , Z — and its individual values with lowercase letters.

Example 1: Roll a fair six-sided die

Let X = the number rolled. The possible values are 1, 2, 3, 4, 5, 6, each equally likely with probability $1/6$.

Example 2: Flip 3 coins

Let X = the number of heads. The possible values are 0, 1, 2, 3. Even though the outcome of each flip is unknown, X is a well-defined number every time we do the experiment.

Why we call it "random"

The value depends on chance. Before the experiment we cannot know what X will be — only what values it *could* take and how likely each value is.

Why it matters

Random variables let us apply arithmetic — sums, averages, comparisons — to random processes. Every AP Statistics topic that follows in this unit (probability distributions, expected value, variance, binomial, geometric, sampling distributions) starts with a random variable.

Practice — What Is a Random Variable?

Q1. A spinner shows 1, 2, 3, 4. Let X = the number the spinner lands on. Is X a random variable? What are its possible values?

Answer:

Q2. A bag has 3 red and 2 blue marbles. You draw one and let $X = 1$ if red, 0 if blue. List the possible values of X .

Answer:

Q3. You flip a coin 4 times and record the number of tails. Let X = number of tails. What values can X take?

Answer:

Q4. True or false: a random variable must be a whole number.

Answer:

Q5. A student says “the color of a marble is a random variable.” Is that correct? Why?

Answer:

2. Discrete vs. Continuous Random Variables



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Discrete vs. Continuous Random Variables

A random variable is a numerical value determined by the outcome of a random process. It can be either **discrete** or **continuous**.

Discrete Random Variable

Takes countable values.

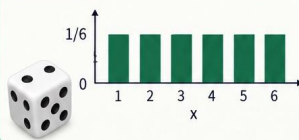
- A discrete random variable can only take **countable values** (finite or infinite).
- The possible values can be listed: 0, 1, 2, 3, ...

Examples:

- Number on a die (1, 2, 3, 4, 5, 6)
- Number of heads in 3 coin flips (0, 1, 2, 3)
- Number of students absent in a class (0, 1, 2, ...)
- Number of goals in a soccer game (0, 1, 2, ...)

Example: Number on a die

x	1	2	3	4	5	6
P(X = x)	1/6	1/6	1/6	1/6	1/6	1/6



Continuous Random Variable

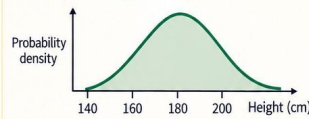
Takes any value in an interval.

- A continuous random variable can take **any value in an interval** (a continuum of values).
- The possible values form a continuum (infinitely many values).

Examples:

- Height of a person (e.g., 150.5 cm, 150.6 cm, ...)
- Time to complete a test (e.g., 23.4 minutes)
- Temperature on a given day (e.g., 72.3°F)
- Weight of a package (e.g., 1.27 kg)

Example: Height of a person



A continuous random variable can take **any value**, such as 165.3 cm, 165.31 cm, 165.312 cm, and so on.

Discrete random variables **count**.
Continuous random variables **measure**.

Two kinds of random variables

Every random variable is either **discrete** or **continuous**. Which kind it is decides how we write its probability model and how we compute probabilities.

Discrete — you can count

A discrete random variable takes **countable** values (finite or countably infinite). The possible values can be listed: 0, 1, 2, 3, ...

- Number on a die: 1, 2, 3, 4, 5, 6
- Number of heads in 3 coin flips: 0, 1, 2, 3
- Number of students absent: 0, 1, 2, ...
- Number of goals in a soccer game: 0, 1, 2, ...

Continuous — you *measure*

A continuous random variable can take **any value in an interval**. The possible values form a continuum — infinitely many values, and between any two you can always find another.

- Height of a person: 150.5 cm, 150.6 cm, 150.63 cm, ...
- Time to complete a test: 23.4 minutes
- Temperature: 72.3°F
- Weight of a package: 1.27 kg

Bar chart vs density curve

Discrete distributions are shown with a **bar chart** (or histogram): one bar per possible value, height = probability. Continuous distributions are shown with a **density curve**: probability is **area** under the curve, not the height at a single point.

Rule of thumb. If the natural question is “How many?” the variable is discrete. If it’s “How much?” the variable is continuous.

Practice — Discrete vs. Continuous Random Variables

Q1. Classify as discrete or continuous: number of cars that pass a light in an hour.

Answer:

Q2. Classify as discrete or continuous: exact time (in seconds) it takes a runner to finish a race.

Answer:

Q3. Classify as discrete or continuous: number of siblings a student has.

Answer:

Q4. Classify as discrete or continuous: the weight of a randomly selected apple.

Answer:

Q5. A random variable takes values 0, 0.5, 1, 1.5, 2 with given probabilities. Is it discrete or continuous?

Answer:

3. Probability Distribution of a Discrete RV



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Probability Distribution of a Discrete Random Variable

A **probability distribution** of a **discrete** random variable shows each possible value of X and the probability that X takes that value.

Create tables showing each possible value of X and its corresponding probability.

What It Shows

- The possible values of the random variable X .
- The probability for each value.
- Written as $P(X = x)$.

Since X is discrete, its values can be listed.

How to Make a Probability Distribution

- 1 Identify the random variable.
- 2 List all possible values of X .
- 3 Find the probability of each value.
- 4 Check that all probabilities add up to 1.

Example 1: Flip 2 Fair Coins

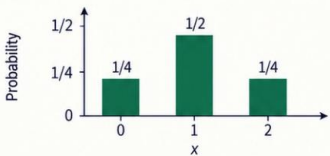
Let X = number of heads.

Possible outcomes:
HH, HT, TH, TT

Possible values of X : 0, 1, 2

Probability distribution:

x	0	1	2
$P(X = x)$	1/4	1/2	1/4



Example 2: Roll a Fair Die

Let X = number rolled.

Possible values of X :
1, 2, 3, 4, 5, 6

Probability distribution:

x	1	2	3	4	5	6
$P(X = x)$	1/6	1/6	1/6	1/6	1/6	1/6



Valid Probability Distribution Rules

- Each probability must be between 0 and 1.
- The sum of all probabilities must equal 1.

$$\sum P(X = x) = 1$$

A discrete probability distribution lists every possible value of X and how likely each value is.

What a probability distribution shows

A **probability distribution** of a discrete random variable X lists every possible value x together with its probability $P(X = x)$. It answers two questions at once: *what values can X take?* and *how likely is each?*

How to build a distribution — four steps

1. Identify the random variable in words. (“Let $X = \dots$ ”)
2. List all possible values of X .
3. Find the probability of each value.
4. Check that all probabilities add up to 1.

Example 1: Flip 2 fair coins, X = number of heads

Sample space: HH, HT, TH, TT (4 equally likely outcomes). X can be 0, 1, or 2.

x	0	1	2
$P(X = x)$	1/4	1/2	1/4

Example 2: Roll a fair die, X = number rolled

x	1	2	3	4	5	6
$P(X = x)$	1/6	1/6	1/6	1/6	1/6	1/6

Valid probability distribution rules

- Each probability must satisfy $0 \leq P(X = x) \leq 1$.
- The probabilities must sum to 1: $\sum P(X = x) = 1$.

Practice — Probability Distribution of a Discrete RV

Q1. Roll a fair die. Write the probability distribution of X = number rolled.

Answer:

Q2. Flip 2 fair coins. Let X = number of tails. Write the probability distribution.

Answer:

Q3. A random variable X has distribution $P(X = 1) = 0.2$, $P(X = 2) = 0.5$, $P(X = 3) = 0.3$. Compute $P(X \geq 2)$.

Answer:

Q4. For the same X , compute $P(X < 3)$.

Answer:

Q5. List the four steps to build a probability distribution for a discrete random variable.

Answer:

4. Rules for a Valid Probability Distribution



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Rules for a Valid Probability Distribution

A probability distribution for a discrete random variable lists every possible value of X and its corresponding probability. For it to be valid, it must satisfy **two rules**.

Rule 1

Each probability must be **between 0 and 1** (inclusive).

$$0 \leq P(X = x) \leq 1$$

for every value of x .

Rule 2

The sum of all probabilities must be equal to 1.

$$\sum P(X = x) = 1$$

(over all possible values of x)

Valid Probability Distribution

x	$P(X = x)$
0	0.10
1	0.30
2	0.40
3	0.20

✓ All probabilities are between 0 and 1.

✓ $0.10 + 0.30 + 0.40 + 0.20 = 1.00$

So this is a **valid** probability distribution.

Invalid Probability Distribution

x	$P(X = x)$
0	0.20
1	0.50
2	0.40
3	-0.10

✗ $P(X = 3) = -0.10$ is not between 0 and 1.

✗ $0.20 + 0.50 + 0.40 - 0.10 = 1.00$ (but Rule 1 is already violated).

So this is **NOT** a valid probability distribution.



Key Takeaway

For a probability distribution to be valid:

- 1 Every probability must be between 0 and 1.
- 2 The probabilities must sum to 1.

Two rules — both must hold

A discrete probability distribution is **valid** only when it passes both checks. Miss either one and the table is not a probability distribution.

Rule 1 — each probability is between 0 and 1

$0 \leq P(X = x) \leq 1$ for every value of x . A negative probability is impossible; a probability greater than 1 is impossible.

Rule 2 — the probabilities sum to 1

$\sum P(X = x) = 1$ summed over every possible value of X . If they sum to more or less than 1, some probability is wrong or a value is missing.

Valid example

x	0	1	2	3
$P(X = x)$	0.10	0.30	0.40	0.20

All four values are in $[0, 1]$ and $0.10 + 0.30 + 0.40 + 0.20 = 1.00$. **Valid**.

Invalid example

x	0	1	2	3
P(X = x)	0.20	0.50	0.40	-0.10

$P(X = 3) = -0.10$ is not between 0 and 1. Even though $0.20 + 0.50 + 0.40 - 0.10$ happens to equal 1.00, Rule 1 is already violated. **Not valid.**

Check Rule 1 first. A negative probability makes the table invalid on the spot, even before the sum works out.

Practice — Rules for a Valid Probability Distribution

Q1. Is this a valid distribution? x: 0, 1, 2; P: 0.3, 0.5, 0.2.

Answer:

Q2. Is this a valid distribution? x: 1, 2, 3; P: 0.4, 0.5, 0.2.

Answer:

Q3. Is this a valid distribution? x: 0, 1, 2, 3; P: 0.25, 0.25, 0.25, 0.25.

Answer:

Q4. Is this a valid distribution? x: 1, 2, 3, 4; P: 0.3, 0.3, -0.1, 0.5.

Answer:

Q5. What are the two rules for a valid discrete probability distribution?

Answer:

5. Finding Missing Probabilities



Finding Missing Probabilities

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The probabilities in a distribution must add up to 1.
Use this fact to find an unknown probability.

$$\sum P(X = x) = 1$$

(over all possible values of x)

Example 1 Find a Missing Probability

The table shows the probability distribution for a random variable X . Find $P(X = 2)$.

x	0	1	2	3
$P(X = x)$	0.20	0.30	?	0.10

Solution:

- 1 Use the fact that the probabilities sum to 1.
 $0.20 + 0.30 + P(X = 2) + 0.10 = 1$
- 2 Add the known probabilities.
 $0.60 + P(X = 2) = 1$
- 3 Solve for the missing probability.
 $P(X = 2) = 1 - 0.60 = 0.40$

Answer: $P(X = 2) = 0.40$

Example 2 Find a Missing Probability

The table shows the probability distribution for a random variable Y . Find $P(Y = 4)$.

y	1	2	3	4
$P(Y = y)$	0.15	0.25	0.35	?

Solution:

- 1 Use the fact that the probabilities sum to 1.
 $0.15 + 0.25 + 0.35 + P(Y = 4) = 1$
- 2 Add the known probabilities.
 $0.75 + P(Y = 4) = 1$
- 3 Solve for the missing probability.
 $P(Y = 4) = 1 - 0.75 = 0.25$

Answer: $P(Y = 4) = 0.25$



Key Takeaway

To find a missing probability in a distribution, use the fact that the sum of all probabilities is 1.

Idea

The probabilities in a distribution must sum to 1. If one probability is missing, use that fact to solve for it: $\sum P(X = x) = 1$.

Method (three steps)

1. Set up the equation $\sum P(X = x) = 1$ with the missing probability as an unknown.
2. Add the known probabilities.
3. Solve for the missing probability.

Example 1: Find $P(X = 2)$

x	0	1	2	3
$P(X = x)$	0.20	0.30	?	0.10

$$0.20 + 0.30 + P(X = 2) + 0.10 = 1 \Rightarrow 0.60 + P(X = 2) = 1 \Rightarrow P(X = 2) = 0.40.$$

Example 2: Find $P(Y = 4)$

y	1	2	3	4
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$P(Y = y)$ 0.15 0.25 0.35 ?

$0.15 + 0.25 + 0.35 + P(Y = 4) = 1 \Rightarrow 0.75 + P(Y = 4) = 1 \Rightarrow P(Y = 4) = 0.25$.

Sanity check. After you fill in the missing value, add the whole column. If it isn't exactly 1, the table has more than one error.

Practice — Finding Missing Probabilities

Q1. x : 0, 1, 2, 3; P : 0.1, 0.4, ?, 0.2. Find $P(X = 2)$.

Answer:

Q2. y : 1, 2, 3; P : 0.15, ?, 0.55. Find $P(Y = 2)$.

Answer:

Q3. x : 0, 1, 2, 3, 4; P : 0.05, 0.15, 0.30, ?, 0.10. Find $P(X = 3)$.

Answer:

Q4. A student solves and gets a missing probability of -0.05 . What went wrong?

Answer:

Q5. z : 1, 2, 3, 4; P : 0.2, 0.2, ?, ?. If $P(Z = 3) = 2 \cdot P(Z = 4)$, find both missing values.

Answer:

6. Expected Value — The Mean of a Random Variable



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Expected Value The Mean of a Random Variable

The expected value of a random variable is the **long-run average outcome** if we repeat the random process many times.

$$\mu_X = E(X) = \sum x P(X = x)$$

Multiply each possible value by its probability, then add the products.

Example 1: Fair Die

Let X be the number that appears on a fair six-sided die.

x	1	2	3	4	5	6
$P(X = x)$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$

Compute the expected value:

$$\begin{aligned} E(X) &= 1\left(\frac{1}{6}\right) + 2\left(\frac{1}{6}\right) + 3\left(\frac{1}{6}\right) + 4\left(\frac{1}{6}\right) + 5\left(\frac{1}{6}\right) + 6\left(\frac{1}{6}\right) \\ &= \frac{1 + 2 + 3 + 4 + 5 + 6}{6} \\ &= \frac{21}{6} = 3.5 \end{aligned}$$



So, the expected value is 3.5.

Example 2: Game Winnings

You play a game that costs \$2 to play. You win \$5 with probability 0.3 and win \$0 with probability 0.7. Let X be your net winnings (in dollars).

x	-2	3
$P(X = x)$	0.7	0.3

Compute the expected value:

$$\begin{aligned} E(X) &= (-2)(0.7) + 3(0.3) \\ &= -1.4 + 0.9 \\ &= -0.5 \end{aligned}$$

The expected value is -0.5.
In the long run, you expect to lose \$0.50 per play.



Key Takeaways

- Expected value is the long-run average outcome.
- Use $E(X) = \sum xP(X = x)$.
- It does not have to be a value that can actually occur.
- It can represent an average result, such as a typical gain, loss, cost, or number of successes.

What expected value is

The **expected value** of a random variable is the **long-run average outcome** if we repeat the random process many times. It is also called the **mean** of the random variable, μ_X .

Formula

$$\mu_X = E(X) = \sum x \cdot P(X = x)$$

Multiply each possible value by its probability, then add up the products.

Example 1: Fair six-sided die

x	1	2	3	4	5	6
$P(X = x)$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$

$$E(X) = 1(1/6) + 2(1/6) + 3(1/6) + 4(1/6) + 5(1/6) + 6(1/6) = 21/6 = 3.5$$

You cannot roll a 3.5, but if you roll a die a million times, the average of the results is very close to 3.5.

Example 2: A \$2 game

The game costs \$2 to play. You win \$5 with probability 0.3 and win \$0 with probability 0.7. Let X be your net winnings in dollars.

x	-2	3
$P(X = x)$	0.7	0.3

$E(X) = (-2)(0.7) + 3(0.3) = -1.4 + 0.9 = -\0.50 . In the long run you expect to lose \$0.50 per play.

Key takeaways

- Expected value is the **long-run** average outcome.
- Use $E(X) = \sum x \cdot P(X = x)$.
- It does not have to be a value X can actually take (e.g. 3.5 on a die).
- It can represent an average gain, loss, cost, or number of successes.

Practice — Expected Value — The Mean of a Random Variable

Q1. x : 0, 1, 2; P : 0.2, 0.5, 0.3. Find $E(X)$.

Answer:

Q2. A spinner has values 1, 2, 5 with probabilities 0.5, 0.3, 0.2. Find $E(X)$.

Answer:

Q3. A game pays \$10 with probability 0.1 and \$0 with probability 0.9. The ticket costs \$2. What is your expected net gain?

Answer:

Q4. A random variable takes values 100, 200, 300 with probabilities 0.5, 0.3, 0.2. Find $E(X)$.

Answer:

Q5. An insurance policy pays \$500 with probability 0.02 and \$0 with probability 0.98. What is the fair (break-even) premium?

Answer:

7. Interpreting Expected Value



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Interpreting Expected Value

The expected value is the **long-run average outcome** if we repeat the random process many times. It represents the average result, even though it **does not have to be a possible individual outcome**.

Example: Number of Heads in 3 Coin Flips

Let X be the number of heads when flipping a fair coin 3 times.

x	$P(X = x)$
0	$1/8$
1	$3/8$
2	$3/8$
3	$1/8$



Each flip is fair, with $P(\text{Heads}) = 1/2$.

Compute the expected value:

$$\begin{aligned} E(X) &= 0\left(\frac{1}{8}\right) + 1\left(\frac{3}{8}\right) + 2\left(\frac{3}{8}\right) + 3\left(\frac{1}{8}\right) \\ &= \frac{0 + 3 + 6 + 3}{8} \\ &= \frac{12}{8} = 1.5 \end{aligned}$$

The expected value is 1.5.

On average, you get 1.5 heads in 3 coin flips.

Important Points

- 1 Expected value is the long-run average outcome.
- 2 It is a weighted average of all possible values, using their probabilities.
- 3 The expected value **does not have to be a possible individual outcome**.

What Does 1.5 Mean?

You cannot get 1.5 heads in a single set of 3 coin flips, but if you repeat this experiment many times, the average number of heads per set will be close to 1.5.

Expected value describes the long-run average, not a single result.



Key Takeaway

Expected value helps us understand the **typical or average outcome** of a random process, even when that value cannot actually occur in a single trial.

What expected value really means

Expected value is the **weighted average** of all possible outcomes, weighted by their probabilities. It tells us what to expect *on average* over many repetitions — **not** what to expect on a single trial.

Example: 3 fair coin flips, X = number of heads

x	0	1	2	3
$P(X = x)$	$1/8$	$3/8$	$3/8$	$1/8$

$$E(X) = 0(1/8) + 1(3/8) + 2(3/8) + 3(1/8) = 12/8 = 1.5$$

What does 1.5 heads mean?

You cannot actually get 1.5 heads in one set of 3 flips (you get 0, 1, 2, or 3). But if you repeat the “flip 3 coins” experiment many times, the **average** number of heads per set will be very close to 1.5.

Three important points

1. Expected value is the **long-run average outcome** — not the most likely value.

2. It is a **weighted average**: more likely outcomes pull the answer more.
3. It **does not have to be a possible outcome**. 1.5 heads, 3.5 on a die, and $-\$0.50$ on a game are all valid expected values you can never actually observe on one trial.

Common mistake. Students hear “expected” and think *most likely*. They’re not the same. The most likely number of heads in 3 flips is 1 or 2 (both have probability $3/8$), but the expected value is 1.5.

Practice — Interpreting Expected Value

Q1. A student computes $E(X) = 2.7$ for the number of heads in 4 flips. Is 2.7 a possible outcome?

Answer:

Q2. What is the difference between the expected value and the most likely value of a random variable?

Answer:

Q3. A shop’s daily profit has $E(X) = \$450$. On a random day, will they earn exactly \$450?

Answer:

Q4. For X = number of heads in 3 fair coin flips, we found $E(X) = 1.5$. Explain in one sentence what 1.5 means.

Answer:

Q5. True or false: $E(X)$ is always one of the possible values of X .

Answer:

8. Expected Value in Games and Decisions

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Good
Decisions
Add Up

Expected Value in Games and Decisions

Calculate expected winnings, losses, profits, or costs to decide whether a game or decision is favorable.

$$E(X) = \sum xP(X = x)$$

Multiply each possible outcome by its probability, then add the products.

How to Interpret It

- 1 Positive expected value: favorable gain or profit.
- 2 Negative expected value: expected loss or extra cost.
- 3 Compare expected values to choose the better option.


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Example 1: Carnival Game

It costs \$2 to play. You win \$10 with probability 0.15, or win \$0 with probability 0.85.

x (net outcome in dollars)	8	-2
$P(X = x)$	0.15	0.85

$$\begin{aligned} E(X) &= 8(0.15) + (-2)(0.85) \\ &= 1.20 - 1.70 \\ &= -0.50 \end{aligned}$$

Expected value = -\$0.50 per play.
This game is not favorable for the player.

Example 2: Buying Shipping Insurance

A package has a 5% chance of being lost, which would cost \$80. Insurance costs \$3.

Option A: No insurance	Option B: Buy insurance
Expected cost: $E(\text{cost}) = 80(0.05) + 0(0.95)$ $= 4$ Expected cost = \$4	Guaranteed cost = \$3 Expected cost = \$3

Since \$3 < \$4, buying insurance is the better expected-value decision.

 **Key Takeaway**
Expected value helps us make smart decisions by comparing the long-run average gain, loss, profit, or cost.

Choose the option with the better expected value.

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Why expected value matters for decisions

Expected value gives us a single number to compare two random options — an insurance policy vs. going uninsured, playing a game vs. sitting out, choosing route A vs. route B. Whichever choice has the better expected value is, on average, the better long-run decision.

How to interpret the sign

- **Positive** expected value: favorable — a long-run gain or profit.
- **Negative** expected value: unfavorable — a long-run loss or extra cost.
- To choose between two options, compute $E(X)$ for each and pick the better one.

Example 1: Carnival game

It costs \$2 to play. Win \$10 with probability 0.15, or \$0 with probability 0.85. Net outcome X :

x (net dollars)	8	-2
$P(X = x)$	0.15	0.85

$E(X) = 8(0.15) + (-2)(0.85) = 1.20 - 1.70 = -\0.50 per play. The game is not favorable — over many plays you lose \$0.50 each time.

Example 2: Buying shipping insurance

A package has a 5% chance of being lost (which would cost you \$80). Insurance costs \$3.

- **Option A — No insurance:** $E(\text{cost}) = 80(0.05) + 0(0.95) = \4.00 .
- **Option B — Buy insurance:** guaranteed cost = \$3.00 .

$\$3 < \4 , so **buying insurance is the better expected-value decision.**

Note. Expected value is a long-run guide. On any single shipment you might feel silly paying \$3 for insurance you didn't need, or you might be very glad you did.

Practice — Expected Value in Games and Decisions

Q1. Game A: win \$5 with prob 0.3, lose \$2 with prob 0.7. Compute $E(X)$ and decide if it's favorable.

Answer:

Q2. Game B: win \$10 with prob 0.15, lose \$3 with prob 0.85. Compute $E(X)$ and decide.

Answer:

Q3. A phone has a 3% chance of breaking (costing \$400 to repair). Warranty costs \$20. Should you buy it (by expected value)?

Answer:

Q4. Now suppose the phone has a 10% chance of breaking. Redo the decision.

Answer:

Q5. Two commutes: A always takes 30 min. B takes 20 min with prob 0.6 and 45 min with prob 0.4. Which has the shorter expected time?

Answer:

9. Variance of a Random Variable



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Variance of a Random Variable

The **variance** measures how much the possible values of a random variable typically **vary around its expected value**.

$$\sigma_X^2 = \sum (x - \mu_X)^2 P(X = x)$$

It is the expected value of the squared distance from the mean.

Example: Fair Die

Let X be the number that appears on a fair six-sided die.

x	$P(X = x)$	$x - \mu$	$(x - \mu)^2$	$(x - \mu)^2 P(X = x)$
1	1/6	-2.5	6.25	1.0417
2	1/6	-1.5	2.25	0.3750
3	1/6	-0.5	0.25	0.0417
4	1/6	0.5	0.25	0.0417
5	1/6	1.5	2.25	0.3750
6	1/6	2.5	6.25	1.0417

1 Find the mean.

$$\mu_X = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = 3.5$$

2 Compute the variance.

$$\begin{aligned} \sigma_X^2 &= 1.0417 + 0.3750 + 0.0417 \\ &\quad + 0.0417 + 0.3750 + 1.0417 \\ &= 2.9167 \end{aligned}$$

3 (Optional) Find the standard deviation.

$$\sigma_X = \sqrt{2.9167} \approx 1.71$$

The variance is 2.9167, and the standard deviation is approximately 1.71.



What Does Variance Tell Us?

- Variance measures the typical squared distance from the mean.
- A larger variance means the values are more spread out.
- A smaller variance means the values are more concentrated around the mean.
- Variance is always nonnegative.

Example: Compare Two Games

Two games have the same expected value (\$0), but different variances.

Game	Outcomes	Probabilities	Expected Value	Variance
A	+\$10, -\$10	0.5, 0.5	0	100
B	+\$2, -\$2	0.5, 0.5	0	4

Both games have the same mean (0), but Game A has a larger variance, so its outcomes are more spread out and less predictable.



Key Takeaways

- 1 Variance measures how much the values typically vary around the mean.
- 2 $\sigma_X^2 = \sum (x - \mu_X)^2 P(X = x)$.
- 3 A larger variance means more variability; a smaller variance means less.

What variance measures

Variance measures how much the possible values of a random variable typically **vary around its expected value**. It is the expected value of the squared distance from the mean.

Formula

$$\sigma_X^2 = \sum (x - \mu_X)^2 \cdot P(X = x)$$

Method (three steps)

1. Find the mean $\mu_X = E(X)$.
2. For each value, compute $(x - \mu_X)^2 \cdot P(X = x)$.
3. Add all the terms.

Example: Fair six-sided die

$$\mu = 3.5$$

$$x \quad P(X = x) \quad x - \mu \quad (x - \mu)^2 \quad (x - \mu)^2 P(X = x)$$

1	1/6	-2.5	6.25	1.0417
2	1/6	-1.5	2.25	0.3750
3	1/6	-0.5	0.25	0.0417
4	1/6	0.5	0.25	0.0417
5	1/6	1.5	2.25	0.3750
6	1/6	2.5	6.25	1.0417

Sum: $\sigma_X^2 = 1.0417 + 0.3750 + 0.0417 + 0.0417 + 0.3750 + 1.0417 = 2.9167$.

Comparing two games with the same mean

Game A pays +\$10 or -\$10 with equal probability. Game B pays +\$2 or -\$2 with equal probability. Both have $E(X) = 0$, but Game A has variance 100 and Game B has variance 4. Same expected value, very different risk.

Key facts

- Variance is always **nonnegative**.
- Larger variance → more spread out, less predictable.
- Smaller variance → more concentrated near the mean.

Practice — Variance of a Random Variable

Q1. x : 0, 1, 2; P : 0.2, 0.5, 0.3. Compute $E(X)$, then $\text{Var}(X)$.

Answer:

Q2. A random variable is constant (always equals 5). What is its variance?

Answer:

Q3. Game A pays $\pm\$10$ with equal probability. Game B pays $\pm\$2$ with equal probability. Which has the larger variance and why?

Answer:

Q4. Can variance be negative?

Answer:

Q5. x : 1, 3; P : 0.5, 0.5. Compute $\text{Var}(X)$.

Answer:

10. Standard Deviation of a Random Variable



Standard Deviation of a Random Variable

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The **standard deviation** is the **square root of the variance**. It measures the typical distance between the values of the random variable and its expected value.

$$\sigma_X = \sqrt{\sigma_X^2}$$

Example 1: Fair Die

Let X be the number that appears on a fair six-sided die.

x	$P(X = x)$	$(x - \mu)^2$	$(x - \mu)^2 P(X = x)$
1	1/6	6.25	1.0417
2	1/6	2.25	0.3750
3	1/6	0.25	0.0417
4	1/6	0.25	0.0417
5	1/6	2.25	0.3750
6	1/6	6.25	1.0417

1 Find the mean.

$$\mu_X = \frac{1 + 2 + 3 + 4 + 5 + 6}{6} = 3.5$$

2 Find the variance.

$$\sigma_X^2 = 2.9167 \text{ (from previous example)}$$

3 Find the standard deviation.

$$\sigma_X = \sqrt{2.9167} \approx 1.71$$

The standard deviation is approximately 1.71, which means the outcomes typically differ from the mean by about 1.71.



Example 2: Coin Tosses (3 Flips)

Let Y be the number of heads in 3 fair coin flips.

y	$P(Y = y)$	$(y - \mu)^2$	$(y - \mu)^2 P(Y = y)$
0	1/8	2.25	0.2813
1	3/8	0.25	0.0938
2	3/8	0.25	0.0938
3	1/8	2.25	0.2813

1 Find the mean.

$$\mu_Y = 0\left(\frac{1}{8}\right) + 1\left(\frac{3}{8}\right) + 2\left(\frac{3}{8}\right) + 3\left(\frac{1}{8}\right) = 1.5$$

2 Find the variance.

$$\sigma_Y^2 = 0.2813 + 0.0938 + 0.0938 + 0.2813 = 0.75$$

3 Find the standard deviation.

$$\sigma_Y = \sqrt{0.75} \approx 0.87$$

The standard deviation is approximately 0.87, which means the number of heads typically differs from the mean by about 0.87.



Key Takeaways

- 1 The standard deviation is the square root of the variance: $\sigma_X = \sqrt{\sigma_X^2}$.
- 2 It measures how much the values typically vary around the mean.
- 3 A larger standard deviation means more variability; a smaller standard deviation means less variability.

What standard deviation measures

The **standard deviation** is the **square root of the variance**. It measures the typical distance between the values of the random variable and its expected value — in the *same units* as the variable itself.

Formula

$$\sigma_X = \sqrt{(\sigma_X^2)}$$

Why we need it in addition to variance

Variance is in squared units (dollars squared, points squared) and is hard to interpret directly. Standard deviation puts the answer back in the original units, so it's the number we actually report.

Example 1: Fair six-sided die

From the previous section, $\sigma_X^2 = 2.9167$. So $\sigma_X = \sqrt{2.9167} \approx 1.71$. On average, an individual roll differs from the mean 3.5 by about 1.71.

Example 2: Number of heads in 3 fair coin flips

$$\mu_Y = 1.5$$

y **P(Y = y)** **(y - μ)²** **(y - μ)² P(Y = y)**

0 1/8 2.25 0.2813

1 3/8 0.25 0.0938

2 3/8 0.25 0.0938

3 1/8 2.25 0.2813

$$\sigma_Y^2 = 0.75, \text{ so } \sigma_Y = \sqrt{0.75} \approx \mathbf{0.87} \text{ heads.}$$

Key takeaways

- $\sigma_X = \sqrt{(\sigma_X^2)}$ — same units as the variable.
- Larger σ means more variability; smaller means less.
- Always nonnegative.

Practice — Standard Deviation of a Random Variable

Q1. $\text{Var}(X) = 16$. Find σ_X .

Answer:

Q2. $\text{Var}(X) = 0.49$. Find σ_X .

Answer:

Q3. Standard deviation is 2.5. Find variance.

Answer:

Q4. For X = number of heads in 3 fair coin flips, $\text{Var}(X) = 0.75$. Find σ_X .

Answer:

Q5. Why do we report standard deviation more often than variance in practice?

Answer:

11. Transforming Random Variables



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Transforming Random Variables

If we **add or multiply** a random variable by constants, the mean and standard deviation change in a **predictable** way.

If $Y = a + bX$, then

$$\mu_Y = a + b\mu_X \quad \text{and} \quad \sigma_Y = |b|\sigma_X$$

Adding a constant (a) shifts the mean but does not change the standard deviation. Multiplying by a constant (b) multiplies the mean by b and the standard deviation by $|b|$.

Example 1: Adding a Constant

Let X be the number on a fair six-sided die. Then $\mu_X = 3.5$ and $\sigma_X \approx 1.71$. Let $Y = X + 4$. Find μ_Y and σ_Y .

1 Use the formulas.

$$\begin{aligned}\mu_Y &= 4 + \mu_X = 4 + 3.5 = 7.5 \\ \sigma_Y &= |1|\sigma_X = 1(1.71) = 1.71\end{aligned}$$

2 Interpret the result.

Adding 4 shifts all outcomes up by 4. The mean increases by 4, but the standard deviation stays the same.

Answer: $\mu_Y = 7.5$, $\sigma_Y = 1.71$

Example 2: Multiplying by a Constant

Let X be the number on a fair six-sided die. Then $\mu_X = 3.5$ and $\sigma_X \approx 1.71$. Let $Y = 2X$. Find μ_Y and σ_Y .

1 Use the formulas.

$$\begin{aligned}\mu_Y &= 2\mu_X = 2(3.5) = 7.0 \\ \sigma_Y &= |2|\sigma_X = 2(1.71) = 3.42\end{aligned}$$

2 Interpret the result.

Multiplying by 2 doubles all outcomes. Both the mean and the standard deviation are multiplied by 2.

Answer: $\mu_Y = 7.0$, $\sigma_Y = 3.42$



Key Takeaways

- 1 If $Y = a + bX$, then $\mu_Y = a + b\mu_X$ and $\sigma_Y = |b|\sigma_X$.
- 2 Adding a constant a shifts the mean but does not change the standard deviation.
- 3 Multiplying by a constant b multiplies the mean by b and the standard deviation by $|b|$.
- 4 The standard deviation is always nonnegative, so we use $|b|$.

Linear transformations change the center and spread in simple, predictable ways.

What happens when you shift or scale a random variable?

If we add or multiply a random variable by *constants*, the mean and standard deviation change in a very predictable way.

Set $Y = a + bX$. Then:

$$\mu_Y = a + b\mu_X \quad \text{and} \quad \sigma_Y = |b|\sigma_X$$

The variance scales by b^2 : $\sigma_Y^2 = b^2 \cdot \sigma_X^2$.

Adding a constant (a shift)

Adding a constant slides every outcome up or down by the same amount. The **mean shifts by a** , and the **standard deviation does not change** — the spread is unchanged.

Multiplying by a constant (a scale)

Multiplying by b stretches or shrinks every outcome. The **mean is multiplied by b** , and the **standard deviation is multiplied by $|b|$** (we use absolute value because standard deviation is always nonnegative).

Example 1: Adding a constant

Fair die: $\mu_X = 3.5$, $\sigma_X \approx 1.71$. Let $Y = X + 4$. Then $\mu_Y = 4 + 3.5 = 7.5$, and $\sigma_Y = 1 \cdot 1.71 = 1.71$.

Example 2: Multiplying by a constant

Same die. Let $Y = 2X$. Then $\mu_Y = 2(3.5) = 7.0$, and $\sigma_Y = 2(1.71) = 3.42$.

Big idea. Adding a constant moves the center. Multiplying by a constant changes both the center and the spread. Neither operation changes the *shape* of the distribution.

Practice — Transforming Random Variables

Q1. $\mu_X = 10$, $\sigma_X = 3$. Let $Y = X + 7$. Find μ_Y and σ_Y .

Answer:

Q2. $\mu_X = 10$, $\sigma_X = 3$. Let $Y = 4X$. Find μ_Y and σ_Y .

Answer:

Q3. $\mu_X = 20$, $\sigma_X = 5$. Let $Y = 3 + 2X$. Find μ_Y and σ_Y .

Answer:

Q4. $\mu_X = 50$, $\sigma_X = 8$. Let $Y = -X + 100$. Find μ_Y and σ_Y .

Answer:

Q5. A student computes $\sigma_Y = -6$ after multiplying by -2 . Why is that wrong?

Answer:

12. Combining Random Variables



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Combining Random Variables

We can find the **mean** and standard deviation of sums and differences of random variables. If the random variables are **independent**, the **variances add**.

General Formulas

Let X and Y be random variables and let a and b be constants. For $Z = aX + bY$:

$$\mu_Z = a\mu_X + b\mu_Y$$

If X and Y are independent:

$$\sigma_Z = \sqrt{a^2\sigma_X^2 + b^2\sigma_Y^2}$$

If X and Y are not independent:

$$\sigma_Z = \sqrt{a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\sigma_{XY}}$$

where σ_{XY} is the covariance of X and Y .

Special Cases

- **Sum (independent):** $Z = X + Y$

$$\mu_Z = \mu_X + \mu_Y$$

$$\sigma_Z = \sqrt{\sigma_X^2 + \sigma_Y^2}$$

- **Difference (independent):** $Z = X - Y$

$$\mu_Z = \mu_X - \mu_Y$$

$$\sigma_Z = \sqrt{\sigma_X^2 + \sigma_Y^2}$$

- **More generally (independent):** $Z = aX + bY$

$$\mu_Z = a\mu_X + b\mu_Y$$

$$\sigma_Z = \sqrt{a^2\sigma_X^2 + b^2\sigma_Y^2}$$

Example 1: Sum of Two Independent Dice

Let X and Y be the outcomes of two independent fair six-sided dice. Let $Z = X + Y$.

Each die: $\mu_X = \mu_Y = 3.5$ $\sigma_X = \sigma_Y \approx 1.71$

Mean of Z : $\mu_Z = 3.5 + 3.5 = 7$

Standard deviation of Z : $\sigma_Z = \sqrt{1.71^2 + 1.71^2} = \sqrt{5.8336} \approx 2.42$

Answer: $\mu_Z = 7$, $\sigma_Z \approx 2.42$



Example 2: Difference (Independent)

Let X and Y be the outcomes of two independent fair six-sided dice. Let $Z = X - Y$.

Each die: $\mu_X = \mu_Y = 3.5$ $\sigma_X = \sigma_Y \approx 1.71$

Mean of Z : $\mu_Z = 3.5 - 3.5 = 0$

Standard deviation of Z : $\sigma_Z = \sqrt{1.71^2 + 1.71^2} = \sqrt{5.8336} \approx 2.42$

Answer: $\mu_Z = 0$, $\sigma_Z \approx 2.42$

What if the Variables Are Not Independent?

If X and Y are not independent, we must include the covariance term:

$$\sigma_Z = \sqrt{\sigma_X^2 + \sigma_Y^2 + 2\sigma_{XY}}$$

(for $Z = X + Y$)

Positive covariance increases the variance.
Negative covariance decreases the variance.

If $\sigma_{XY} = 0$ (independence), the formula reduces to $\sigma_Z = \sqrt{\sigma_X^2 + \sigma_Y^2}$.



Key Takeaways

- 1 The mean of a sum or difference is the sum or difference of the means.
- 2 If the random variables are independent, the variances add.
- 3 If they are not independent, include the covariance term.
- 4 Independence is important when combining variances.
- 5 These rules also work for constants a and b : $Z = aX + bY$.

Combining random variables helps us analyze total outcomes, differences, and linear combinations in many real-life situations.

Sums and differences of random variables

We often want to know the mean and standard deviation of a **combination** of random variables — the total of two dice, the difference of two test scores, the total time of two independent tasks. The rules for the mean are simple; the rules for standard deviation depend on whether the variables are **independent**.

Mean of a linear combination (always true)

For $Z = aX + bY$ with any constants a and b :

$$\mu_Z = a\mu_X + b\mu_Y$$

Special cases: $\mu_{X+Y} = \mu_X + \mu_Y$ and $\mu_{X-Y} = \mu_X - \mu_Y$.

Variance rule (only if independent)

If X and Y are **independent**, the **variances add**, even for a difference:

$$\sigma_Z^2 = a^2\sigma_X^2 + b^2\sigma_Y^2$$

So $\sigma_Z = \sqrt{a^2\sigma_X^2 + b^2\sigma_Y^2}$. Note that **standard deviations do NOT add** — you must square first.

Example 1: Sum of two independent dice

Each die: $\mu = 3.5$, $\sigma \approx 1.71$. For $Z = X + Y$: $\mu_Z = 7$ and $\sigma_Z = \sqrt{1.71^2 + 1.71^2} = \sqrt{5.8336} \approx 2.42$.

Example 2: Difference of two independent dice

Same dice. For $Z = X - Y$: $\mu_Z = 3.5 - 3.5 = 0$, but variances still add: $\sigma_Z = \sqrt{1.71^2 + 1.71^2} \approx 2.42$.

What if X and Y are NOT independent?

Include the covariance term: $\sigma_Z^2 = a^2\sigma_X^2 + b^2\sigma_Y^2 + 2ab\sigma_{XY}$. Positive covariance *increases* the variance of a sum; negative covariance *decreases* it.

Test-day rule. Means always add or subtract. Variances only add if the variables are independent — and even for differences, variances still *add* (they never subtract).

Practice — Combining Random Variables

Q1. $\mu_X = 10$, $\mu_Y = 4$. Find μ_{X+Y} and μ_{X-Y} .

Answer:

Q2. $\sigma_X = 3$, $\sigma_Y = 4$, X and Y independent. Find σ_{X+Y} .

Answer:

Q3. $\sigma_X = 3$, $\sigma_Y = 4$, X and Y independent. Find σ_{X-Y} .

Answer:

Q4. Two independent fair dice. Find the mean and standard deviation of their sum.

Answer:

Q5. A student writes $\sigma_{X+Y} = \sigma_X + \sigma_Y$ for independent X, Y. Correct?

Answer: